

ASSIGNMENT

Subject: Generative AI -25-CAT-673

25MAM-1 A & B

Submission Date: 30th Jan., 2025

MARKS: 40 Marks

1. Explain Sequential Data Modeling and mention two real-world applications.
2. What are Recurrent Neural Networks (RNNs)? How are they used for next-word prediction?
3. Why do LSTMs perform better than standard RNNs for long sequences?
4. Explain the Encoder–Decoder architecture with one example application.
5. What are sequence-to-sequence (Seq2Seq) models? Mention any two tasks they solve.
6. Differentiate between Natural Language Generation (NLG) and Natural Language Understanding (NLU).
7. Explain the role of deep learning models in text generation.
8. What are multilingual language models? Name any two examples.
9. Explain mBERT and its importance in cross-lingual tasks.
10. What is XLM-R? How is it different from monolingual models?
11. What is self-attention? Why is it important in transformer models?
12. Explain multi-head attention and state one advantage.
13. Briefly explain the architecture of a Transformer model.
14. Compare BERT and GPT in terms of architecture and usage.
15. Explain any one tokenization technique (BPE / WordPiece / SentencePiece) with a suitable example.
16. What is tokenization in NLP? Briefly explain word-level and character-level tokenization.
17. Define subword tokenization. Why is it preferred over word-level tokenization?
18. What is Byte Pair Encoding (BPE)? How does it help in reducing the OOV problem?
19. What is WordPiece tokenization? Explain the role of the ## symbol with an example.
20. What is SentencePiece? Why is it suitable for morphologically rich or space-less languages?